

Week 1 Progress Report

Project: Log File Analyzer for Intrusion Detection

1. Introduction

During the first week of the internship, the primary focus was on setting up the development environment and understanding the structure of system log files. The objective was to build a foundation for analyzing logs and extracting meaningful information for intrusion detection.

2. Objectives Achieved

- 1 Set up Kali Linux environment and Python workspace
- 2 Installed required libraries such as pandas, matplotlib, and requests
- 3 Collected Apache and SSH log files for analysis
- 4 Designed initial project structure with modular Python files

3. Technical Work

The parsing module was developed to extract key information from raw logs using regular expressions. Apache logs were processed to retrieve IP addresses, timestamps, HTTP methods, URLs, and status codes. SSH logs were analyzed to capture failed login attempts. The extracted data was structured into pandas DataFrames for further processing.

4. Tools and Technologies Used

- 1 Python 3
- 2 pandas for data handling
- 3 matplotlib for visualization (planned)
- 4 Regex for log parsing
- 5 Kali Linux environment

5. Outcome

By the end of Week 1, a functional log parsing system was established. Raw log files could be converted into structured data, enabling further analysis in subsequent phases. This laid the groundwork for implementing intrusion detection techniques in the upcoming weeks.

6. Conclusion

Week 1 successfully established the foundation of the project by focusing on environment setup and data extraction. The system is now ready for the implementation of detection algorithms in the next phase.